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Beneficiation and applications of an Egyptian bentonite

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Abstract

Samples of Egyptian bentonitic sediments, representing Gabal Hamdal (Miocene) of Abu-Zeneima locality, were evaluated by different techniques for some applications such as bleaching, refining of oils and drilling mud. Mineralogically the samples were characterized as low grade of Ca–Mg smectite. The dominant gangue mineral is calcite with accessory amounts of kaolinite, quartz, feldspar and dolomite. Characterization of these samples indicated that the clay response for Na-exchange was poor while their hydration, plastic and rheological properties were inferior to those of commercial bentonites. The samples were processed by crushing, attrition scrubbing followed by hydrocycloning to give pre-concentrates which were leached with diluted HCl. Application of such simple flowsheet effectively removed the majority of the associated calcite, dolomite, feldspar and quartz. These concentrates were further refined by a series of acid and alkaline treatments prior to their evaluation as industrial bentonites. The mineralogy, chemical analysis and physical properties of final concentrates and their activated products successfully match the required specifications for many industrial processes. The bentonite concentrates after alkaline activation are suitable for drilling muds while the products of acid activation can be used as bleaching clay. Cost–benefit analysis of the suggested flowsheet, of beneficiation and activation, proved the economic viability of such processes. Utilization of these clays, after their upgrading and activation, can represent a value-added to the Egyptian economy by preventing the importation of high quality activated clays. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: bentonite; smectite; Fuller's earth; mineralogy; beneficiation; activation

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1. Introduction

Montmorillonite is by far the most abundant of the smectite clay minerals. It is the predominant mineral composing bentonite as well as fuller's earths throughout the world and is a major component of soils formed in arid and semiarid areas. In industry, montmorillonite are generally classified as sodium (Na) or calcium (Ca) types, depending on which exchangeable ion is dominant. The vast majority of montmorillonite deposits outside the US are the calcium type. Sodium montmorillonite is, however, known to occur in Mexico, Argentina, South Africa, Turkey, Japan, and Australia (Odom, 1987).

In the US, acid-activated clays are produced almost exclusively from southern or non-swelling bentonites by treatment with various mineral acids. The first commercial acid-activated clays were produced in Germany about 1905. Filtrol, the first US producer, began its operations in 1922. In the intervening years, plants have been built in England, Canada, Italy, Spain, India, Japan, and later in Mexico, Brazil, and China (Taylor and Jenkins, 1987). At present, the western world has an estimated capacity of about 550 kt/a. Obviously, these tonnages are not large in comparison to the total worldwide bentonite production, estimated at about 6 Mt/a. Typical acid-activated bleaching clay prices generally fall in the range of US\$200–400/t, and more specialized products such as catalysts and color developing clays for carbonless copy paper coatings can be even higher (Taylor and Jenkins, 1987).

Bentonites consist essentially of clay minerals of the smectite group and has a wide range of industrial applications. The major problems facing the utilization of the Egyptian bentonites are their low concentration of smectite, high level of impurities, and inconsistent composition.

Previous studies on some Egyptian bentonites (such as Qasr El-Sagha, Kom-Osheim, Maadi, Aun-Musa, etc.) suggested that, without beneficiation, they were unsuitable for most industrial applications (drilling mud, foundry, refining and bleaching, etc.). Response of these bentonitic clays to Na-exchange was poor and their hydration, plastic and rheological properties were inferior to those of standard commercial bentonites (Felix, 1977; Ibrahim, 1988; Ingelthorpe et al., 1993; Ibrahim et al., 1994). Several attempts were made to upgrade these low grade bentonites to meet OCMA/API specifications for drilling fluid and other industrial uses (CMRDI, 1983; GPC report, 1984; Boulos et al., 1988; Rasheed and Amer, 1996).

Currently, Central Metallurgical Research and Development Institute (CMRDI), Egypt, produces on a pilot plant scale, through a research project with Sinai Manganese, Egypt, both the acidic and alkaline activated bentonite. They use bentonites of Qasr El-Sagha locality. These ores are not subjected to any beneficiation process, to separate their associated gangue minerals (mainly gypsum), but the company depends on selective mining. The economic viability of these activation processes is proven.

The Nukhul Formation (Miocene) of Abu-Zeneima (Sinai, Egypt) is, also, a bentonite-bearing sediment. This clay contains typical divalent-cation saturated smectite, traces of kaolinite and quartz in addition to large amounts of calcite. The presence of such impurities hinder its application in oil clarification as some

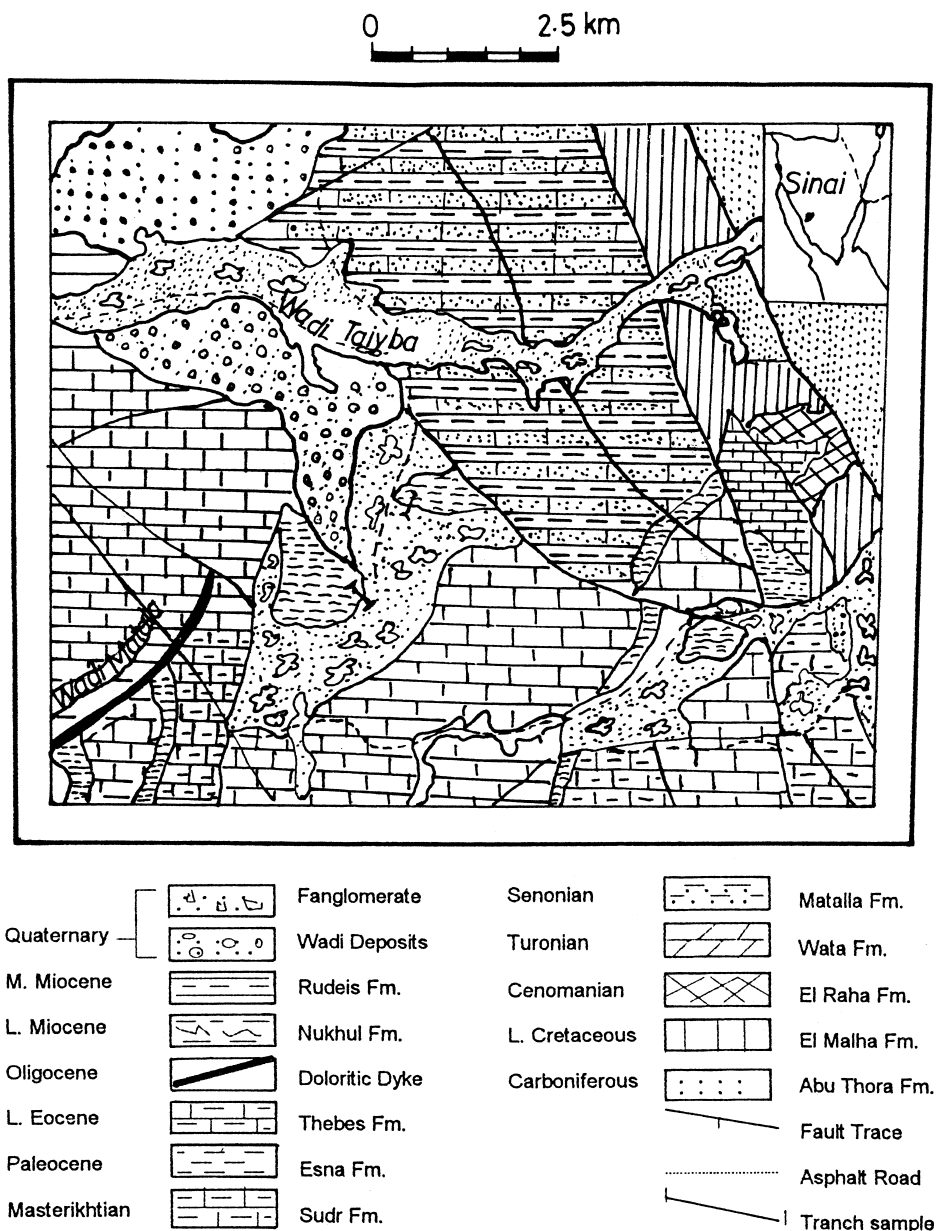


Fig. 1. Geological map of Abu-Zeneima area (Sinai).

of these minerals neutralize the acid used to activate the clay. Also, the low concentration of smectite (less than 50% Ca-smectite) makes it sub-standard for uses such as drilling mud.

This paper presents an evaluation for the Gabal Hamdal bentonitic clay, of Abu-Zenemia (Sinai), for application in drilling mud, refining and bleaching of oils. The tests were conducted on run-of-mine (ROM), concentrates, and chemically activated samples.

1.1. Geology

The samples used in this study represent an area located some 10 km northeast of Abu-Zeneima on the left side of Wadi Taiyba (Sinai). The stratigraphic sequence in the area starts from the carboniferous sandstones of Abu-Thora Formation (carboniferous age) to the lower Miocene calcareous sandstones and shale of Rudeis Formation (M. Miocene age) as shown in the geological map (Kora, 1984), Fig. 1.

The Nukhul Formation (Lower Miocene age) is bentonite bearing. It is well-exposed in the area as a separate mesa covered with Quaternary conglomerates, and in other parts as the base of the calcareous sandstones of the Rudeis Formation (M. Miocene age). The latter exhibits great facies changes but is generally made up of calcareous sandstones with oyster shell bands, pecten-bearing limestone, shale, clay and polymictic conglomerate of Eocene chert nodules. The Nukhul Formation (Lower Miocene age) is proposed to represent shallow marine deposits, and it is of Burdigaliam (Miocene) age (Said, 1990).

Six trench samples were collected and measured at the Gabal Hamdal locality (Sinai). There were two bentonite clay beds with thickness from 1.5 to 3 m each and an over burden from 0.5 to 2 m of Quaternary deposits.

2. Experimental

2.1. Methods of samples characterization

All trench samples were dried at 55°C and then crushed using a 'Wedag' jaw crusher to 100% below 5 mm in size. A representative batch of each sample was further ground to 100% below 200 mesh using a porcelain mill to avoid contamination of samples. These samples were used for characterization and physico-chemical tests.

Characterization of samples was performed using X-ray diffraction (bulk and oriented samples below 2 μm), chemical analysis and thermal analysis. Procedures of the characterization and physico-chemical tests followed those of Wilson (1987), Inglethorpe et al. (1993), and Carter et al. (1965). The surface

area of each sample was determined using a technique which measures the quantity of ethylene glycol monoethyl ether (a volatile polar organic liquid) adsorbed by a known weight of powdered sample under vacuum. The method assumes that the reagent is adsorbed, under vacuum, as a monolayer on the surface of silicate minerals (Carter et al., 1965). Cation exchange capacity (CEC) of the samples was determined by a methylene blue absorption technique (Taylor, 1985).

Swelling indices were determined by a standard technique of measuring the gel volume formed from the addition of 1 g, air dried, sample in 10 ml of water after 24 h when the last portion of the material was introduced into the cylinder. Also, the swelling indices were measured for the samples after the addition of Na_2CO_3 (1–5% by weight), which is the common practice in bentonite activation (Inglethorpe et al., 1993).

Testing of the samples as a raw material for drilling fluid was carried out according to Section 11 of the API specifications (American Petroleum Institute, 1990).

Testing of such samples for clarification of vegetable oils was carried out after acid activation. For assessment of the ability of the samples in the decolorizing of the vegetable oils, 0.25 g of the activated product was thoroughly mixed with 20 ml of sun flower oil for 30 min in an oil bath at a bleaching temperature of 100°C. The bleaching ability was, then, determined by measuring the color of bleached oil using a Lovibond Tintometer with a 5.25" cell. The yellow and red colors were recorded.

2.2. Rheological tests

In testing the rheological properties of the activated products, the suspension were re-agitated for 5 min, using a mixer, and then transferred to the viscometer at a rotor speed of 600 rpm. The readings of the viscometer were recorded at 5 min intervals until a stable reading was attained. This procedure was repeated at a rotor speed of 300 rpm. The reading of the viscometer at 600 rpm is taken as the apparent viscosity (A.V.) while the difference between the readings taken at 600 and 300 rpm represents the plastic viscosity (P.V.). The yield point (Y.P.) is calculated from the difference between the reading of viscometer at 300 rpm and plastic viscosity.

2.3. Methods of beneficiation

The bentonite ore was beneficiated to improve its smectite content by removing the associated gangue minerals (mainly calcite and quartz). The sample was primary crushed using a Jaw crusher to 100% below 5 mm in size. The primary crushed samples were first attrition scrubbed at a high solids

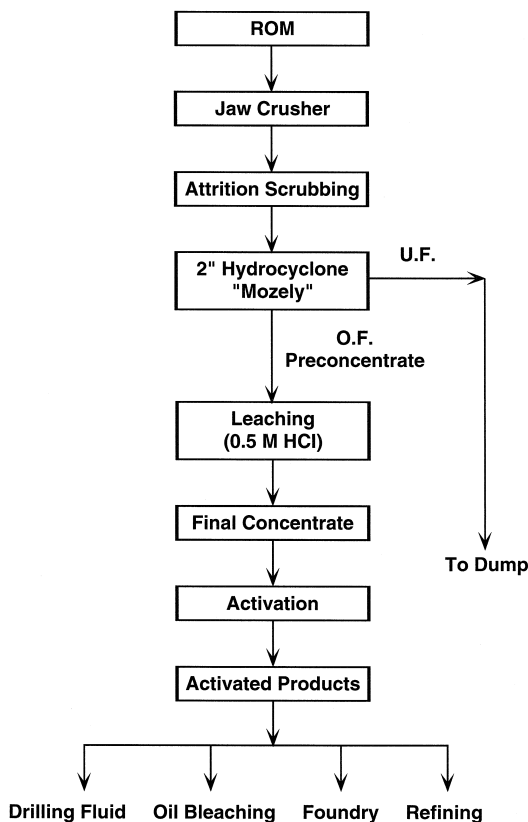


Fig. 2. A flowsheet for the beneficiation scheme of Abu-Zeneima bentonite ores.

concentration (50%) and at high impeller speed (2500 rpm) for 1 h, using a 'Denver D12' flotation cell, as an attritioner, to disintegrate the smectite from its associating gangue minerals. This was followed by classification using a 'Mozley' 2' hydrocyclone at 10% solids and at pressure of 30 psi.

Both the hydrocyclone underflow (coarser than 44 μm) and the overflow fractions (finer than 44 μm) were filtered, dried at 55°C and weighed. The underflow fractions, which were heavily contaminated with quartz and calcite impurities, were discarded. Meanwhile, the overflow fractions were used as bentonite pre-concentrates. The latter products were further upgraded by leaching, their residual calcite impurities, with 0.5 M HCl for 1 h using the flotation cell, as a mixer, at a speed of 2500 rpm. Fig. 2 shows a simplified flowsheet for the beneficiation scheme.

2.4. Methods of activation

In the production of alkaline-activated bentonite, samples (22.5 g) were mixed with (1–5% by weight) Na_2CO_3 . The mixture was added to 350 ml of

distilled water and the suspension was stirred for 20 min before allowing to age overnight (Garrett, 1987; Odom, 1987).

On the other hand, for production of acid-activated bentonite HCl of different strength (2–6 N) for different contact times (up to 7 h) was used for activation, after which the product was washed with hot water to pH 7 and dried (Taylor and Jenkins, 1987).

3. Results and discussion

3.1. Beneficiation of the bentonite ore

The results of beneficiation showed that applying a flowsheet based on attrition scrubbing of the crushed bentonite ore followed by classification with a hydrocyclone succeeded in removing much of the associated quartz and calcite as an underflow (about 15 wt.%) fraction. So, this underflow fraction was discarded. The overflow (about 85 wt.%) from the hydrocyclone was taken as a bentonite pre-concentrate. The chemical analysis of the latter showed that the free silica and calcite contents are significantly decreased in comparison with the feed sample, Table 1.

Meanwhile, leaching of the obtained bentonite pre-concentrate with 0.5 M HCl showed that such technique was successful in removing the majority of the remaining calcite where the CaO content decreased to a minimum value (1.77%). This is reflected on the composition of the smectite mineral where the Al_2O_3 is significantly increased to 15.40% in comparison with 5.6–7.6% in the ROM samples. Also, the contents of SiO_2 , Fe_2O_3 , MgO and TiO_2 increased with increasing smectite content, Table 1.

3.2. Characterization of samples

The bulk X-ray diffraction (XRD) analysis of the ROM samples indicated the presence of calcite (CaCO_3), as the principal non-clay mineral (Fig. 3a). This was also confirmed from the chemical analysis of that with the higher content of CaO (19.7–36.9%) and L.O.I. (20.0–30.8%), Table 1. Quartz and dolomite are also present in minor amounts. XRD analysis of oriented mounts of the ROM samples confirmed the presence of divalent-cation saturated smectite. The basal spacing, d_{001} , of the smectite mineral ranges from 14.45–15.31 Å. Such basal spacing (d_{001}) are expanded to 16.28–16.31 Å with ethylene glycol. Kaolinite is identified in all samples by its basal spacings at 7.14 and 3.56 Å (Fig. 4).

DTA thermograms of the ROM samples (Fig. 5a) reveal low-temperature endothermic peaks at about 138–152°C within which adsorbed and exchange-

Table 1
Chemical analysis of the investigated samples

Samples	SiO ₂ (%)	TiO ₂ (%)	Al ₂ O ₃ (%)	Fe ₂ O ₃ (%)	MgO (%)	CaO (%)	Na ₂ O (%)	K ₂ O (%)	LOI(%)
MJ-48	40.67	0.03	7.64	0.79	0.80	19.69	6.15	0.79	20.0
MJ-50	32.20	0.46	6.60	3.60	0.80	22.84	2.33	0.80	28.2
MJ-51	31.00	0.36	6.50	3.20	1.20	21.28	2.23	0.94	30.8
MJ-52	34.17	0.41	5.60	2.80	1.60	22.96	1.92	0.94	27.1
MJ-53	39.18	0.25	6.10	0.63	0.83	22.72	5.39	0.73	22.0
MJ-56	31.86	0.16	7.24	0.75	0.79	36.94	4.27	0.75	26.0
Pre-concentrate	42.40	0.83	11.62	7.06	1.89	10.5	1.59	0.98	24.0
Final concentrate	51.01	1.15	15.40	8.21	2.44	1.77	0.35	0.77	18.0
Clay fraction	50.40	1.25	15.62	7.81	2.12	1.55	0.36	0.65	18.8

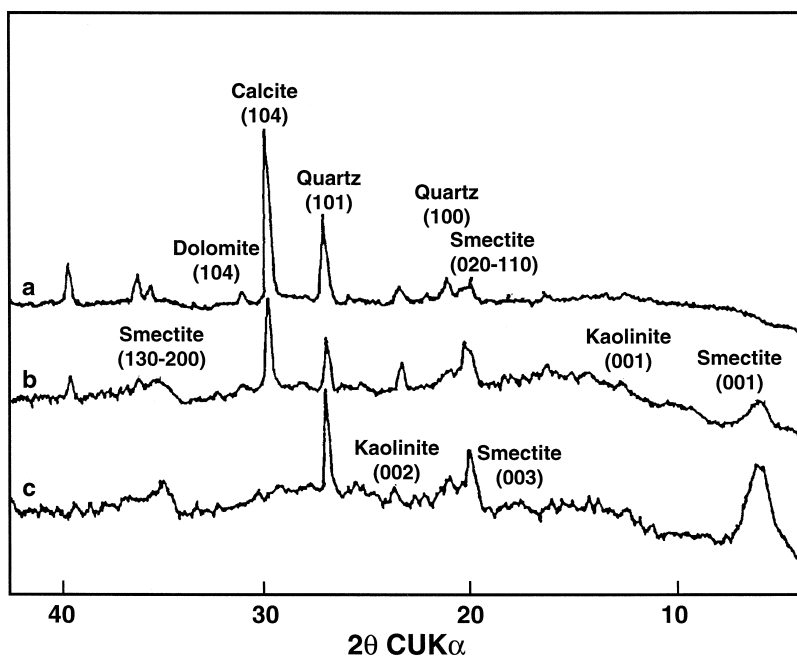
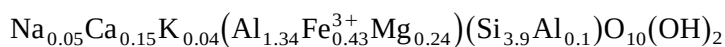


Fig. 3. XRD patterns of the investigated samples. (a) ROM sample. (b) Bentonite pre-concentrate. (c) Final concentrate.

able-cation coordinate water are released. Dual endothermic peaks at 538–566°C and 674–694°C represented dehydroxylation of abnormal smectite. The relatively low peak temperature of dehydroxylation of the samples may be attributed to the presence of excess structural iron in the smectite. These results agreed fairly well with those of Mackenzie and Mitchell (1966). The endothermic peaks at 762 and 862°C correspond to the decomposition of dolomite and calcite, respectively. Exothermic peaks at high temperatures (above 900°C) correspond mainly due to crystallization of new phases, Fig. 5a.

Thermogravimetric curves were used to calculate the different forms of water (water molecules and chemically bound OH-groups). Also, they can be used to present quantitative information about the impurities such as carbonates, Table 2 and Fig. 5.

Table 1 shows the chemical composition of the clay fraction less than 2 μm . Assuming iron is trivalent, the following example demonstrates the average formula of the studied samples.



On the other hand, the XRD data of the obtained bentonite concentrate clearly showed a significant improvement in the smectite content. It indicated very

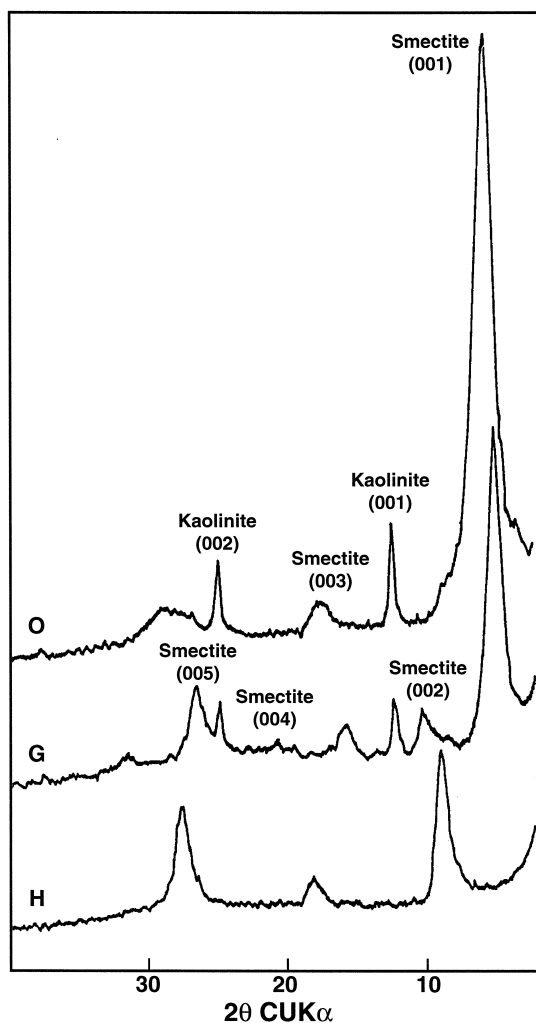


Fig. 4. XRD patterns of the oriented clay fraction.

weak diffractions for quartz and kaolinite but none of calcite and dolomite, Fig. 3c. This suggests that the applied beneficiation flowsheet is succeeded in improving the grade of the bentonite concentrate by removing the majority of its associated gangue minerals. The chemical analysis of the final concentrate becomes very close to that of the clay fraction indicating its high purity, Table 1. This is consistent with the DTA and TG curves (Fig. 5b,c). The intensity and the shape of these peaks are increased as the smectite content increased (Fig. 5b,c). A parallel increase in weight loss is also noticed in the thermogravimetric curves of the obtained concentrate, Table 2.

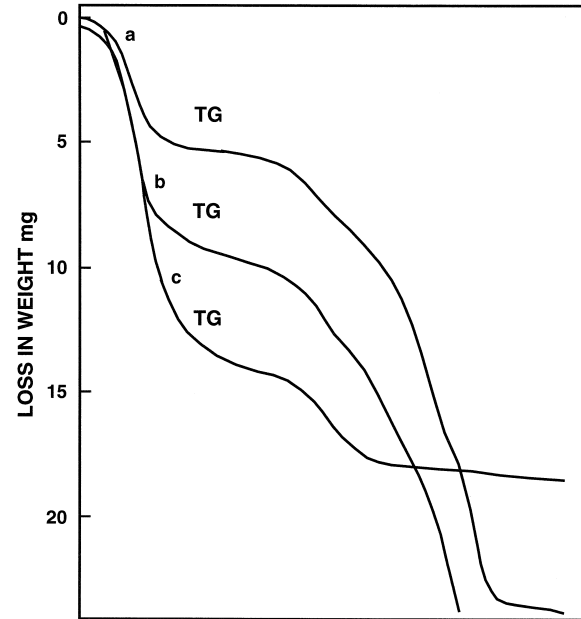
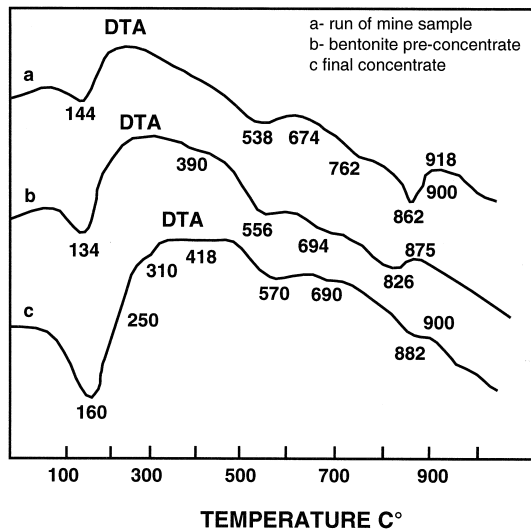


Fig. 5. Thermal analysis of the investigated samples. (a) ROM sample. (b) Bentonite pre-concentrate. (c) Final concentrate.

Table 2
Thermogravimetric data of the investigated samples

Sample	H ₂ O (%)	OH (%)	CO ₂ (%)	LOI (%)	Dehyd. (°C)	Dehx. (°C)	S-shape (°C)		Dec. (°C)		Goeth (°C)
							end.	ex.	do.	ca.	
MJ-48	7.02	5.9	12.37	22.92	148	549–689	880–905		737–864		–
MJ-50	6.29	4.51	17.37	28.16	152	556–691	880–905		761–865		–
MJ-51	6.56	7.83	16.46	30.80	149	550–687	895–925		761–865		–
MJ-52	3.38	4.55	19.12	27.05	138	566–694	880–925		762–862		–
MJ-53	4.82	4.55	15.12	24.49	149	550–700	880–915		761–865		–
MJ-56	5.87	7.85	11.58	25.30	144	538–674	900–918		761–865		–
Pre-conc.	9.10	5.20	10.70	25.00	134	556–694	810–870		–		390
Final conc.	14.02	4.80	–	18.80	160–250	570–690	874–914		–		310–418
Clay frac.	14.50	4.00	–	18.50	159–230	580–680	890–920		–		310–418

Dehyd. = culmination of dehydration.

Dehx. = culmination of dehydroxylation.

S-Shape = endothermic–exothermic peaks.

Dec. = decomposition of dolomite (do.) and calcite (ca.).

Goeth = goethite.

3.3. Technological properties of samples

Determination of the surface area of the smectite group minerals, other clay minerals and the non-clay minerals of the ROM samples showed that they have 800, < 40 and < 5 m²/g, respectively. The large surface area of the sample indicates the presence of smectite. Also, the MB CEC values of the ROM samples range from 40 to 43 meq/100 g, indicating the presence of smectite in all of them. Typical commercial grade bentonite have MB CEC values between 40 and 120 meq/100 g (Taylor, 1985). This means that the MB CEC values of the investigated ROM samples are at the lower end of the desirable range.

The estimated smectite content of the ROM samples are less than 50% and in the past they have been considered of little economic potential as bentonite resources. However, if they are beneficiated they might be useful in some industrial applications.

Table 3 shows the values of the swelling indices of the ROM samples and after their activation with 1–5% Na₂CO₃. The swelling indices of the ROM samples ranged from 20 to 25 which reflect their low grade. After adding 4–5% Na₂CO₃ the maximum values of swelling index were achieved, (Table 3). However, the maximum swelling indices of high grade smectite used in industry are still much greater (as shown in Table 3).

Rheological data of ROM bentonite samples recorded θ_{600} (viscosity at a rotor speed of 600 rpm) which is well below the minimum required value of 30 cP. Incremental addition of Na₂CO₃ resulted in a progressive increase in the values of the apparent viscosity (A.V.) whereas the values of the plastic

Table 3
Swelling index of the investigated samples

Na ₂ CO ₃ (%)	Swelling index										
	Sample MJ-50			Sample MJ-51			Sample MJ-52			HFE	
	A	B	C	A	B	C	A	B	C		
0	22	31	41	22.5	30	40	25	33	42	30	
1	26	60	60	21	48	60	50	65	78	46	
2	55	78	100	33	60	48	63	72	100	74	
3	70	82	123	77	74	120	74	85	125	85	
4	78	95	148	64	90	135	79	90	145	95	
5	70	100	> 150	80	100	151	80	> 100	152	83	

A = ROM bentonite sample.

B = Bentonite pre-concentrate.

C = Final concentrate.

HFE = High Fuller Earth (Standard specification).

viscosity (P.V.) remained relatively constant, (Table 4). Meanwhile, the results of filter loss values of the ROM samples are also given in Table 4. Such results are higher than the maximum standard limit and this could be attributed to both the relatively low smectite content and the presence of high proportions of non-swelling minerals in these ROM samples.

On the other hand, the technological properties of the bentonite concentrate and its activated products are significantly improved after beneficiation. For example, the values of the swelling indices (Table 3) revealed an excellent grade by progressively mixing with different amounts (1–5%) of Na₂CO₃. The swelling indices of activated samples exceeded 100 when 3–5% Na₂CO₃ is added. Testing of Gabal Hamdal bentonite, after its beneficiation, as a possible material for drilling mud surpass the standard properties and this property also substantially exceeds the requirements as shown in Table 4 (American Petroleum Institute, 1990; Taylor and Jenkins, 1987).

At the same time, the plastic viscosity of the activated clay concentrates, varies between 9–10 cP, fall well within the oil grade bentonite requirements (5 cP for Meycogel and 11.0 cP for the distinguished Wyoming bentonites).

Meanwhile, testing of the obtained bentonite concentrate in oil clarification gave excellent results due to the increase of the smectite content upon beneficiation. The bleaching ability of the ROM, pre-concentrate and final concentrate are calculated, the results of which are shown in Table 5. It is clear that there is a sharp increase in the bleaching ability which started from 20% for the ROM sample, to 55% for the pre-concentrate up to > 90% for the final concentrate. The data of acid activation showed that the maximum bleaching ability is achieved after contact with 6 N HCl for 300 min. Fig. 6 shows that the bentonitic deposits under investigation, after their beneficiation, are able, in general, to give bentonite concentrates of the required properties for the market.

Table 4

(A) Specifications of bentonite for drilling fluids

ID	Measurement	OCMA
1	Suspension properties	30 min
	Viscometer dial reading at 600 rpm	
2	Yield point, lb/100 ft ²	3×plastic viscosity max.
3	Filtrate	16.0 ml max.
4	Wet screen analysis	2.5 max.
	Residue on US sieve 200 mesh	
5	Moisture	13 max.

(B) Tests results for the samples treated with 5% Na₂CO₃

ID *	Sample MJ-50			Sample MJ-51			Sample MJ-52			Standard	
	A	B	C	A	B	C	A	B	C	D	E
(1)	15	34	40	15	32	41	20	41	49	45	60
(1a)	9	27	31	9	25	31	13	31	38	34	51
(2)	3	20	22	3	18	21	6	21	27	23	42
(3)	80	22	13	85	24	13	75	16	9	13	6.8
(4)	0.86	–	–	1.5	–	–	0.4	–	–		
(5)	5.7	8.8	13	5.9	8.5	12.5	5.6	8.8	14.2		

* Measurements as in (A).

(1a) = Viscometer dial reading at 300.

A = ROM sample.

B = Bentonite pre-concentrate.

C = Final concentrate bleached with 0.5 M HCl.

D = Cedars type bentonite, Wyoming (Alther, 1986).

E = Thailand bentonite (5% Na₂CO₃).

Table 5

Bleaching ability of the bleached concentrate sample (6 N HCl treatment)

Time (h)	Yellow color	Red color
1	2	0.5
2	1	0.2
3	2	0.5
4	2	0.5
5	1	0.1
6	2	0.5
7	5	0.7
oil	40	3.0
Run-of-mine	32	2.5
Pre-concentrate	24	1.5
Final concentrate	18	0.8

Laboratory scale.

3.4. Cost–benefit analysis

Estimation of the operating cost for the activation processes without and after beneficiation have been conducted using the Egyptian tariffs for mining, transportation, labor, water, power, utilities, etc. The estimated costs for the alkaline-activated clays are about US\$22 and US\$30/t without and after beneficiation, respectively while for the acid-activated clays are about US\$21.2 and US\$27.5/t without and after beneficiation, respectively. This shows clearly that the cost/ton might not significantly increase as a result of applying the proposed beneficiation flowsheet for upgrading these types of Egyptian clays. This is because the beneficiation process will decrease the consumption of acids and alkali used in the activation processes. At the same time, the products, from the technical point of view, will be more effective as indicated before. It should be mentioned that application of the suggested beneficiation and activation

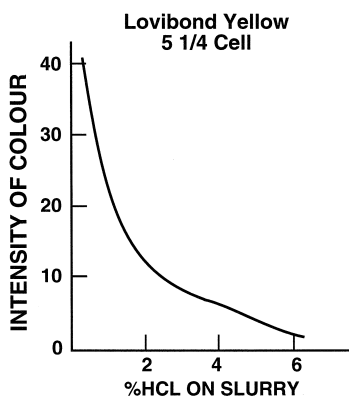


Fig. 6. The bleaching ability of the investigated samples at different HCl concentrations for 300 min.

flowsheet will, indeed, represent a value-added scheme for such Egyptian clays since the cost/ton of the finished products will be about US\$21–30/t in comparison with US\$150–400/t for the other imported bentonites.

4. Conclusions

Evaluation of some samples of Egyptian bentonite-bearing sediments representing Gabal Hamdal (Miocene) of Abu-Zeneima locality (Sinai) indicated that they are of low grade divalent-cation saturated smectite. The estimated smectite content are less than 50%. The main associated gangue mineral is calcite with traces of kaolinite, quartz, feldspar and dolomite. Characterization of such samples indicated that the clay response for Na-exchange is poor and their hydration, plastic and rheological properties are inferior to those of commercial bentonites and in turn they must be beneficiated before being used in different purposes.

Beneficiation of these samples by crushing and attrition scrubbing followed by hydrocycloning gave a hydrocyclone overflow fraction (pre-concentrate) of better grade where most of the associated free silica and calcite are removed. This pre-concentrate is further refined by acid leaching using 0.5 M HCl. Such final concentrate has a minimum CaO content (1.77%). This is reflected on its montmorillonite mineral where the Al_2O_3 is significantly increased to 15.40% in comparison with 5.6–7.6% in the ROM samples.

The mineralogy, chemical analysis and physical properties of the final concentrate and its activated products meet the specification required for application in drilling mud. Meanwhile, testing of the bentonite concentrate in oil clarification gave excellent results due to the increase of the smectite content upon beneficiation. The bleaching ability showed a sharp increase which started from 20% for the ROM samples, to 55% for the pre-concentrate and up to more than 90% for the final concentrate. The data of acid activation showed that the maximum bleaching ability is achieved after contact with 6 N HCl for 300 min. In general, these bentonitic deposits, after their beneficiation and activation, are able to give bentonite concentrates of the required properties for the market.

Cost–benefit analysis of the suggested flowsheet indicated that beneficiation and activation of these Egyptian clays might represent a value-added scheme to the Egyptian economy since the cost/ton of the finished products will be about US\$21–30/t in comparison with US\$150–400/t for the other imported bentonites.

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